AI4EO Final Report

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1 Introduction

Surface-water mapping from satellite imagery is relevant to flood response, reservoir monitoring, and broader hydrological analysis. In optical remote sensing, simple spectral indices such as the Normalized Difference Water Index (NDWI) and the Modified NDWI (MNDWI) remain attractive because they are inexpensive to compute, interpretable, and often surprisingly strong [2, 3, 4]. However, their decision boundaries are fixed and can become brittle under shadows, mixed pixels, turbidity, or non-water dark surfaces. Deep learning has already been applied to flood and water mapping for several years [6, 5], so the contribution of this project is not to claim novelty in the general use of deep learning for water extraction. Instead, the goal is to make a controlled comparison between simple spectral rules, pixel-wise multispectral machine learning, and spatial deep learning in a common Sentinel-2 benchmark setting.

The motivating application for this project is reservoir monitoring in the Tehran region. That application remains relevant, but the contribution is a benchmark study on Sen1Floods11 using hand-labeled Sentinel-2 chips. This narrower focus allows the project to address a methodological question that remained under resolved in the midterm report: when a U-Net appears to outperform simple spectral indices, how much of that gain comes from spatial modeling and how much simply comes from using all Sentinel-2 bands instead of only two bands? To address that question, the final project compares two spectral-index baselines (NDWI and MNDWI), a 13-band Random Forest baseline, and a corrected 13 band U-Net.

The central claim of the final report is therefore modest but useful: on this benchmark, much of the gain over simple spectral indices came from richer multispectral information and nonlinear pixel-wise classification rather than from spatial deep learning alone. This framing avoids overclaiming benchmark performance as validated transfer performance on Tehran reservoirs.

2 Dataset

The experiments use the hand-labeled Sentinel-2 subset of the Sen1Floods11 dataset. Each chip is a 512×512 GeoTIFF containing 13 Sentinel-2 bands, and each corresponding label mask contains pixel values $\{-1, 0, 1\}$, where -1 denotes invalid or no-data pixels, 0 denotes non-water, and 1 denotes water. Sentinel-2 reflectance values were normalized by dividing by 10,000, and invalid pixels were excluded both from loss computation and from evaluation metrics. This subset is well suited to the current project because the final intended application is optical surface-water delineation from Sentinel-2 imagery rather than SAR only flood mapping.

The final experiments used the official hand-labeled split files provided with Sen1Floods11. After matching the split CSVs to the locally downloaded S2Hand and LabelHand files, the resulting benchmark contained 251 training chips, 88 validation chips, and 89 test chips, with no unmatched rows. The valid training pixels were strongly imbalanced toward non-water, which motivated the use of class weighting in the U-Net loss and stratified sampling for the Random Forest baseline.

A brief justification is needed for the choice of Sentinel-2 instead of Sentinel-1. Sen1Floods11 is often associated with flood mapping and mixed S1/S2 analysis. The choice here is motivated by both the final application and the benchmark design. The intended downstream task is reservoir surface-water mapping, for which Sentinel-2 is operationally attractive because it provides directly useful green, NIR, and SWIR information for water discrimination. More importantly for this report, the optical modality enables a meaningful comparison between traditional water indices, classical multispectral machine learning, and deep semantic segmentation within a single consistent input space.

3 Models and Methods

3.1 Spectral baselines

Two classical spectral baselines were evaluated: NDWI and MNDWI. NDWI was computed from the green and NIR bands, and MNDWI from the green and SWIR-1 bands. For each index, a scalar threshold was optimized on the validation set by sweeping a fixed range of thresholds and selecting the threshold that maximized Intersection over Union (IoU). This gives the indices a fair benchmark setting rather than relying on an arbitrary threshold.

3.2 Random Forest baseline

To address the main methodological weakness of the midterm comparison, a 13-band Random Forest classifier was added as a pixel-wise multispectral baseline. The model uses all 13 Sentinel-2 bands as input features but does not exploit spatial context. Training was performed on a stratified sample of valid pixels drawn from the training split, with balanced water and non-water sampling to keep the computation practical. The decision threshold was tuned on the validation set using predicted water probabilities. This baseline is critical because it separates the benefit of richer spectral information and nonlinear decision boundaries from the benefit of convolutional spatial modeling.

3.3 U-Net

The deep learning model is a standard U-Net configured for 13-channel input and binary semantic segmentation. The encoder and decoder architecture uses convolutional blocks with batch normalization and bilinear upsampling. The model was trained with weighted cross-entropy loss so that water pixels received higher loss weight than non-water pixels. Validation IoU was used for model selection, and the best checkpoint was retained.

I discovered a critical implementation issue in the earlier U-Net code: the AdamW optimizer had been re-instantiated inside the batch loop, which prevented optimizer state from being preserved and disconnected the effective updates from the learning rate scheduler. In the final experiments this issue was corrected by creating a single optimizer outside the loop and reusing it throughout training. Because the originally reported U-Net score was obtained with the buggy version, the corrected rerun is the only result used as the final deep-learning benchmark.

3.4 Evaluation

The primary evaluation metric is IoU, with F1 score, precision, recall, and accuracy reported as secondary metrics. All metrics are computed only on valid pixels, excluding mask value -1. IoU is emphasized because it penalizes both missed water and false positives and is standard for segmentation. In addition to aggregate test metrics, qualitative visual comparisons were generated on representative test scenes to inspect boundary quality and typical failure modes.

4 Results

Table 1 summarizes the final comparison. The spectral baselines remained reasonably strong despite their simplicity, with NDWI and MNDWI reaching test IoU values of 0.736 and 0.774, respectively. This confirms that classical water-index heuristics are not trivial baselines to beat in optical water mapping.

The 13-band Random Forest achieved the best overall performance, with a test IoU of 0.849. This result is methodologically important even before considering the deep model, because the Random Forest isolates the effect of using the full Sentinel-2 spectral signature and a nonlinear classifier without any explicit spatial context. Compared with NDWI and MNDWI, it shows that a large part of the gain over simple spectral rules can already be obtained through multispectral pixel-wise learning.

The corrected 13 band U-Net reached a test IoU of 0.759. In this benchmark, the Random Forest outperformed the corrected U-Net, which reinforces the importance of strong classical baselines and cautions against assuming that a convolutional model is automatically superior. The corrected U-Net nevertheless achieved the highest recall (0.966), indicating that it missed little water, but its lower precision (0.780) shows that it also produced more false positives than the Random Forest. In other

words, the U-Net tended to produce broader water masks, whereas the Random Forest achieved the best overall balance between omission and commission errors.

Qualitative examples in Figure 1 are consistent with these quantitative results. NDWI often performs surprisingly well on visually simple scenes, but it can miss parts of complex water structures or behave less robustly near boundaries. The Random Forest generally produces tighter and cleaner masks by leveraging the full multispectral signature. The corrected U-Net often predicts broader connected water regions, which is consistent with its high recall but lower precision. Taken together, the figure and the table suggest that the dominant gain in this benchmark came from richer multispectral information rather than from spatial deep learning alone.

Table 1: Test-set comparison on the final Sen1Floods11 official hand-labeled split.

Model	IoU	F1	Precision	Recall	Accuracy
NDWI	0.736	0.848	0.788	0.918	0.959
MNDWI	0.774	0.873	0.860	0.886	0.967
RF-13	0.849	0.918	0.930	0.907	0.980
UNet-13	0.759	0.863	0.780	0.966	0.961

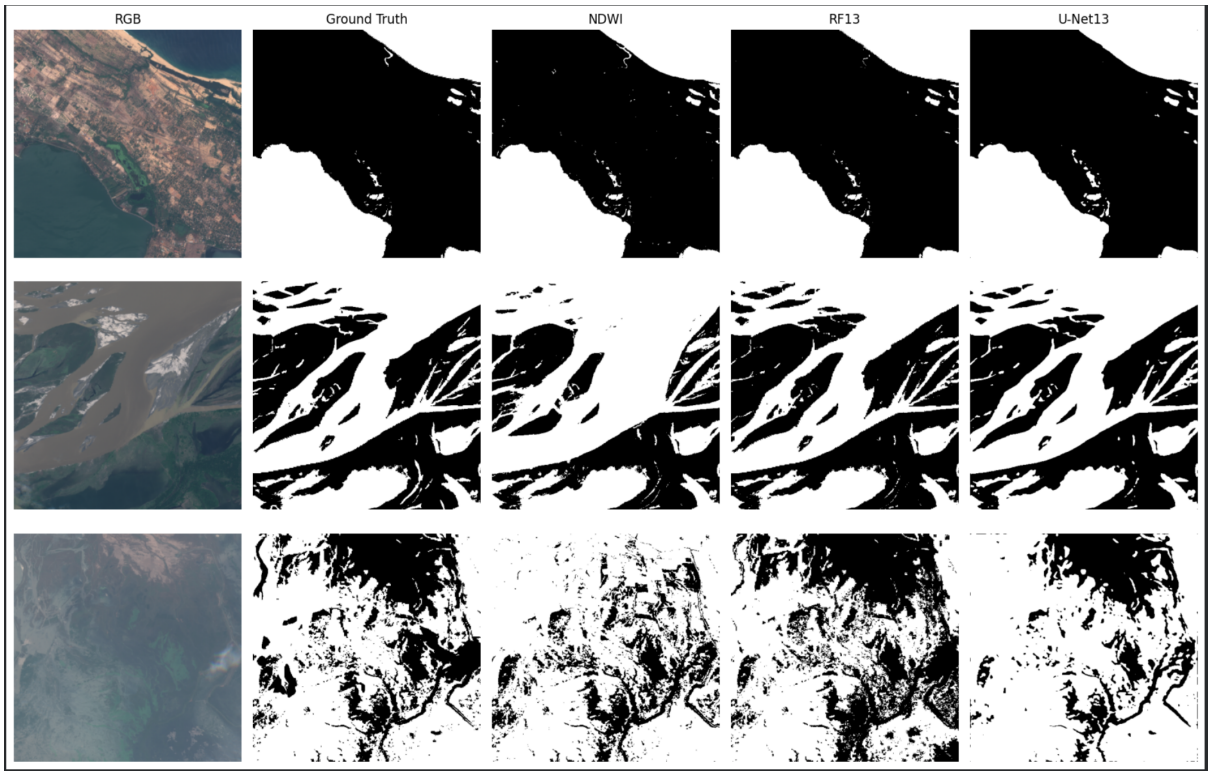

Figure 1: Representative test scenes showing RGB input, ground truth, NDWI prediction, RF-13 prediction, and corrected U-Net prediction.

5 Discussion

The most important difference between the final report and the midterm is that the comparison is now methodologically cleaner. The midterm showed that a 13-band U-Net outperformed two-band spectral indices, but that result alone did not prove that spatial representations were responsible for the improvement. By introducing a 13-band Random Forest, the final benchmark separates two different effects: the benefit of richer multispectral information and the benefit of spatial modeling. The final result shows that, in this setting, richer multispectral information and nonlinear pixel-wise classification mattered more than spatial deep learning.

Several real methodological challenges shaped the final project. First, there was a scope definition challenge: the project initially tried mixing multiple datasets and an ambitious Tehran transfer goal, but the final submission had to be narrowed to a defensible benchmark core. Second, invalid-pixel handling was nontrivial because the Sen1Floods11 labels include a -1 class that must be excluded consistently from both training loss and evaluation; treating those pixels incorrectly would have biased both optimization and metrics. Third, the strong imbalance between water and non-water pixels made naive training potentially misleading, so the final pipeline used class weighting for U-Net and stratified sampling for the Random Forest. Fourth, the original midterm comparison was confounded because it compared two-band spectral indices against a 13-band deep model; adding the 13-band Random Forest was the key methodological fix. Finally, a late-discovered optimizer bug in the earlier U-Net implementation materially changed the interpretation of the deep learning result and required a corrected rerun before the final comparison could be trusted.

Several limitations remain. This is still an in-distribution experiment on Sen1Floods11 rather than a validated Tehran transfer study. The dataset itself focuses on flood-event imagery, whereas the motivating downstream application concerns permanent and seasonal reservoir dynamics, so some domain shift should be expected. In addition, although the final submission uses the official hand-labeled split, benchmark performance on a single dataset still should not be overinterpreted as evidence of broad geographic transferability.

These limitations do not make the benchmark uninformative. On the contrary, they clarify what can and cannot be concluded from the current project. The final submission supports a benchmark claim about the relative behavior of simple indices, multispectral classical ML, and spatial deep learning on handlabeled Sentinel-2 flood-event chips. It does not yet support a strong claim about transfer performance on Tehran reservoirs. The Tehran application is therefore best presented as future work.

A final practical lesson from the project is that implementation correctness matters as much as architectural choice. Discovering the optimizer bug materially changed the interpretation of the earlier U-Net result: after correction, the deep model no longer outperformed the strongest classical baseline. In that sense, the project became less about maximizing a single headline score and more about producing a compact, reproducible, and methodologically defensible Earth-observation benchmark.

sectionReferences

References

[1] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, B. Hoersch, C. Isola, P. Laberinti, P. Martimort, A. Meygret, F. Spoto, O. Sy, F. Marchese, and P. Bargellini, "Sentinel-2: ESA's Optical High-Resolution Mission for GMES Operational Services," Remote Sensing of Environment, vol. 120, pp. 25–36, 2012. doi: 10.1016/j.rse.2011.11.026.

[2] S. K. McFeeters, "The Use of the Normalized Difference Water Index (NDWI) in the Delineation of Open Water Features," International Journal of Remote Sensing, vol. 17, no. 7, pp. 1425–1432, 1996. doi: 10.1080/01431169608948714.

[3] H. Xu, "Modification of Normalised Difference Water Index (NDWI) to Enhance Open Water Features in Remotely Sensed Imagery," International Journal of Remote Sensing, vol. 27, no. 14, pp. 3025–3033, 2006. doi: 10.1080/01431160600589179.

[4] C. Zhao, H. Wei, G. L. Feyisa, T. de Castro Tayer, G. Ma, H. Wu, and Y. Pan, "Evaluating Spectral Indices for Water Extraction: Limitations and Contextual Usage Recommendations," International Journal of Applied Earth Observation and Geoinformation, vol. 139, p. 104510, 2025. doi: 10.1016/j.jag.2025.104510.

[5] D. Bonafilia, B. Tellman, T. Anderson, and E. Issenberg, "Sen1Floods11: A Georeferenced Dataset to Train and Test Deep Learning Flood Algorithms for Sentinel-1," In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 835–845, 2020. doi: 10.1109/CVPRW50498.2020.00113.

[6] X. X. Zhu, D. Tuia, L. Mou, G.-S. Xia, L. Zhang, F. Xu, and F. Fraundorfer, "Deep Learning in Remote Sensing: A Comprehensive Review and List of Resources," IEEE Geoscience and Remote Sensing Magazine, vol. 5, no. 4, pp. 8–36, 2017. doi: 10.1109/MGRS.2017.2762307.

[7] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization," In Proceedings of the International Conference on Learning Representations (ICLR), 2019.

How the project satisfies the grading criteria

1. Breadth

The project covers several methodological families from the course rather than relying on a single model. First, it uses classical remote-sensing feature engineering through NDWI and MNDWI, which directly connects to spectral analysis and optical Earth-observation preprocessing. Second, it uses a classical machine-learning baseline, a 13-band Random Forest, which brings in supervised classification, feature-based learning, and threshold-based decision calibration. Third, it uses a deep semantic-segmentation model, U-Net, which connects to convolutional architectures, pixel-wise prediction, class imbalance handling, and model selection using validation metrics. Fourth, it incorporates core Earth-observation data-handling concepts such as multispectral Sentinel-2 imagery, invalid-pixel masking, official benchmark splits, and geospatial evaluation concerns. Overall, the project combines concepts from remote-sensing preprocessing, classical ML, deep learning, benchmark design, and evaluation rather than focusing narrowly on one lecture topic.

2. Depth

The project goes beyond a simple "indices versus deep learning" comparison by investigating a more specific methodological question: whether performance gains come from spatial modeling or simply from access to richer multispectral information. This was explored by introducing a 13-band Random Forest as an additional control baseline. That comparison was not part of the original midterm and pushed the project beyond a straightforward lecture-level reproduction.

Depth was also added through implementation and evaluation choices. The pipeline explicitly handles invalid pixels labeled as -1 so that they are excluded consistently from both loss computation and metric calculation. The work also addresses strong class imbalance using weighted loss for U-Net and stratified sampling for Random Forest training. In addition, the final project moved from a custom random split toward the official hand-labeled split provided with Sen1Floods11, which improved the rigor of the benchmark design. These steps reflect deeper engagement with dataset structure, evaluation validity, and experimental control than a simple end-to-end training exercise.

3. Challenge

The project was challenging not because it introduced an entirely new architecture, but because it required converting an initially broad and partially confounded project idea into a defensible Earth-observation benchmark. Several aspects contributed to this challenge.

First, the project scope originally mixed multiple possible datasets and an intended Tehran transfer application, so a major challenge was narrowing the work to a realistic and coherent final contribution. Second, the dataset itself introduced nontrivial issues: Sen1Floods11 contains invalid labels that must be masked carefully, and the water class is a minority class, so naive training and evaluation would have been misleading. Third, the original comparison was methodologically weak because it compared two-band spectral indices with a 13-band deep network, which did not isolate the source of improvement. Adding the 13-band Random Forest was a deliberate attempt to fix that confounding factor.

A further challenge was implementation correctness. During the final phase, a bug was identified in the original U-Net training loop: the AdamW optimizer had been recreated inside the batch loop, which invalidated the intended optimizer state and weakened the trustworthiness of the earlier deep-learning result. Correcting and rerunning the model changed the interpretation of the project and made the final benchmark more credible. In this sense, the challenge of the project lay in experimental control, debugging, and methodological refinement rather than in simply reproducing an existing published number.

4. Results

The final results are interesting because they do not support the simplest expected narrative. The project did not end with "deep learning wins." Instead, the strongest model on the final benchmark was the 13-band Random Forest, which achieved an IoU of 0.849, outperforming MNDWI (0.774), the corrected U-Net (0.759), and NDWI (0.736). This is a useful result because it shows that much of the gain over simple spectral indices can already be obtained through richer multispectral pixel-wise classification, even without explicit spatial deep learning.

The corrected U-Net nevertheless achieved the highest recall (0.966), which suggests that it tended to recover more water pixels but also produced more false positives, lowering its precision and IoU relative to the Random Forest. This makes the benchmark more informative than the midterm version: instead of only asking whether a U-Net beats NDWI, the final project reveals how the balance between multispectral information, nonlinear classification, and spatial modeling actually played out on this dataset.

More broadly, the project demonstrates the feasibility of using AI for surface-water segmentation in optical Earth observation, while also showing that strong classical baselines remain highly competitive and should not be omitted. This is a meaningful result for both benchmark design and practical EO model selection.